

Abstract

Kaarel Hänni asked whether a Solomonoff-style inducer over axiom systems, with a prior over templates, a preference for short derivations and a time penalty, would pass from instances $\varphi(t)$ to $\forall x\varphi$, and whether it would robustly recover the axioms we actually have, or at least a deductively equivalent system. We make such an inducer precise. Theories are finite sets of templates with a proper prefix-code prior; the likelihood is a derivation grammar whose normaliser does not depend on the data, which gives the size principle; time is charged for checking axioms or for written derivation length. First, closed instances never favour $\forall x\varphi$ over its instance schema: the odds stay at the prior odds or fall by a constant factor per datum. “All future data are instances” is confirmed, while “the axioms prove $\forall x\varphi$ ” tends to a prior share or to zero, a Bayesian ω -gap. Second, with well-specified data the posterior concentrates on the theories with the data’s law and identifies their theorems, not the axioms. Third, human-written data are not drawn from the model; the posterior then goes to the best-fitting generator, which can be weaker than the axioms behind the data, or unsound; so the actual axioms are not robustly recovered. Fourth, penalising the time to check or generate axioms does not stop Hänni’s collapse of axiom induction into function induction; derivation length in written symbols prices it. All positive guarantees need well-specified data. Adversarial referees with independent code checked the results; no human has.

References